

Corrigendum

Referring to RECOM Asia's advertisement published in Electronics For You magazine, May 2018 on page No. 35, the advertisement contained a typographical error in the first bullet point.

Error: $US\$0.10 \leq \text{OEM Price} \leq US\1.50 (with low MOQ)

Correction: $US\$1.00 \leq \text{OEM Price} \leq US\1.50 (with low MOQ)

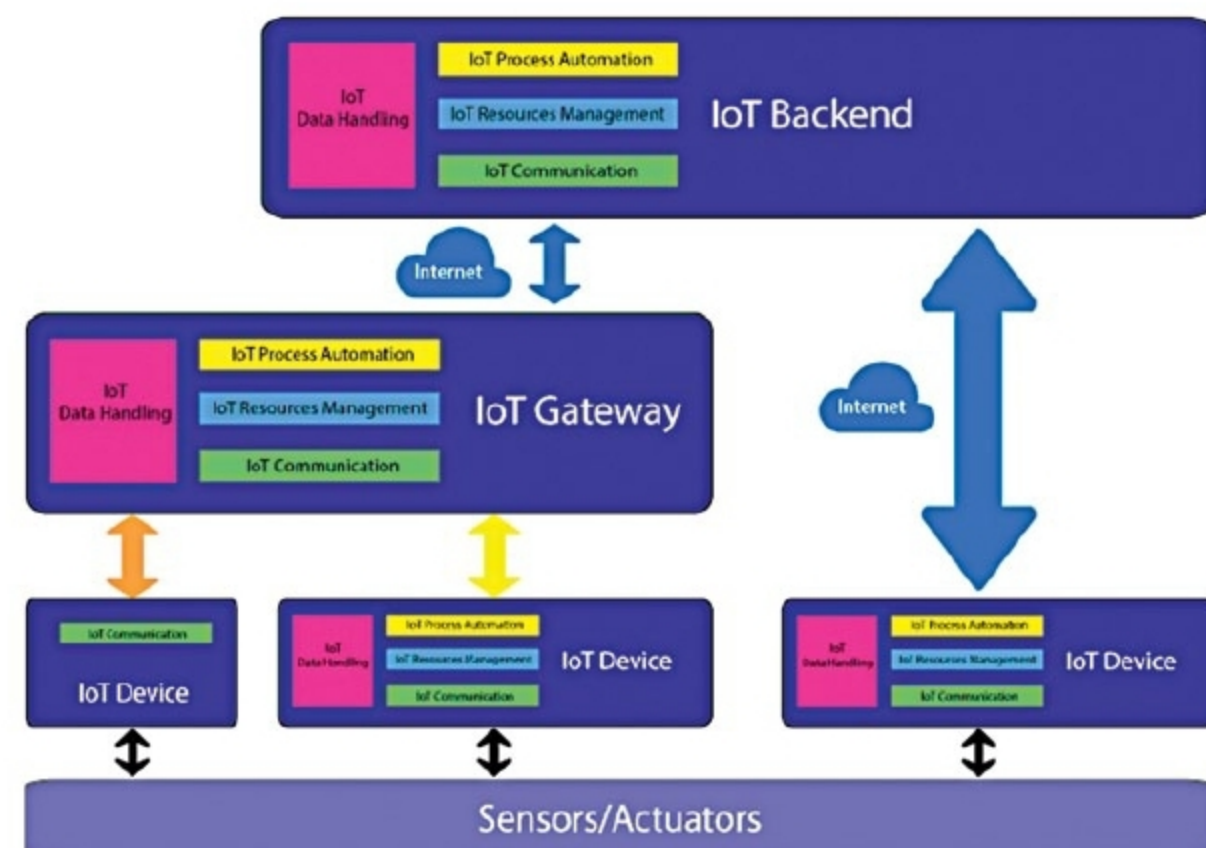
RECOM team regrets the error and sincerely apologises the misrepresentation.

Designs with efficient power consumption to keep devices working through the day are needed. While planning for power for IoT designs it is important to strategise regarding budget. Power budget should be allocated for each functional circuit block on the PCB. This provides a lot more flexibility than just considering power consumption for the product as a whole.

Wireless connectivity

An IoT product may be wired or wireless. It essentially allows a device to sense its surroundings, collect data from multiple sensors and send it to the cloud for interpretation, to present a broader view of the equipment, environment or individual. This is achieved through a variety of wireless, off-the-shelf modules and RF circuits. But, to find room for all of these is a challenge. These parts must have footprints as small as possible while giving all needed functionalities.

Use of wireless modules and protocols in a PCB depends on several factors. These include data trans-



IoT architecture (Credit: www.hindawi.com)

fer speeds, security requirements, required range and power consumption. One or many of these protocols can be used to get the job done—from identification, communication and transport to infrastructure, discovery, data and device management.

There are many other IoT protocols in existence and plenty more in development as the need of our connected future changes. The basic workflow is to build a firewall to protect the IoT devices from unauthorised external access.

Ashish Syal, chief engineer - IoT, at mangOH, says, "Working with customers, we found that a number of designers often do not try new ideas for the IoT because it is either too expensive or takes too long. We wanted to make it

easy for makers and companies to convert their ideas to prototypes, and into products, in the shortest possible time. To do this, we came up with mangOH open hardware platform. It is a highly flexible and expandable open hardware reference design, purpose-built for mobile connectivity. It simplifies the

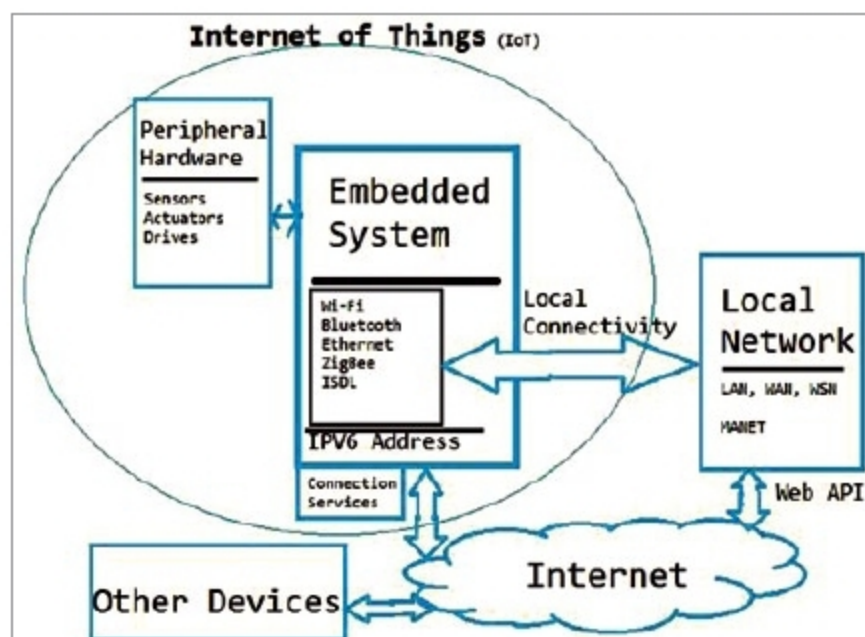
complexity involved in developing new connected mobile products, and removes interoperability issues between mobile and other wired, wireless and sensor technologies."

He adds, "We are targeting IIoT applications that have seen massive adoption in mobile asset tracking, smart agriculture, smart cities, capital asset monitoring and drones. The boards that we are building are provided under the open source creative commons licence attribution style. This is the most accommodating of licences offered, used for maximum dissemination."

There are two core elements of project mangOH. First, a series of open platform designs that you can modify and use to build commercial products. You can deploy these reference designs as is, for prototyping. Or you can use these to build your own platform. Basically, the designs cover 80 per cent of the customer use-cases for the IoT.

Second, IoT expansion cards allow adding different combinations of technology. These cards can be used for prototyping and in actual products. Broken wires or connectors causing major issues can be resolved through it.

Syal comments, "Our latest board is mangOH Red. This platform offers many features, including the capability to build low-power IoT applications that can run for 10 years on a battery, snap-in socket to add any compatible modules



Smart irrigation system using the IoT (Credit: Google Images)